

Scoped literature answer for the polyhedral k -smoothness question

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July 18, 2026

Status

This is a literature-implied answer for a scoped subcase of the final question in Mal–Dey–Paul [1]. It does not answer the arbitrary Banach-space question. It records that the later work of Sain–Sohel–Paul [2] answers the explicitly suggested real finite-dimensional polyhedral case

$$\mathbb{X} = \ell_\infty^n, \quad \mathbb{Y} = \ell_1^n, \quad n \geq 3.$$

Source question

The source paper arXiv:2007.12628 ends the section on k -smoothness of operators defined on Banach spaces with the question: given Banach spaces $\mathbb{X}, \mathbb{Y}$ and $T \in S_{\mathbb{L}(\mathbb{X}, \mathbb{Y})}$, what are necessary and sufficient conditions for T to be a multi-smooth point of finite order? The same question specifically suggests considering $\mathbb{X} = \ell_\infty^n$, $\mathbb{Y} = \ell_1^n$, $n \geq 3$.

Supporting theorem

Sain–Sohel–Paul [2] introduce an index of smoothness $i_R(T)$ for an operator T relative to a subset $R \subset S_{\mathbb{X}}$. Their Definition 1.2 defines this index by taking the dimension of an explicitly described finite-dimensional span of coefficient vectors. Proposition 1.7 proves that the index is independent of the auxiliary bases used in the definition. Theorem 1.8 then proves that, under the standard reflexive/M-ideal hypotheses appearing there,

$$T \text{ is } k\text{-smooth} \iff i_{M_T}(T) = k.$$

In finite dimensions the compact operators are all operators, so the M-ideal and distance hypotheses become trivial. Section II of [2], restricted there to real finite-dimensional polyhedral Banach spaces, gives a three-step finite procedure for determining $i_{M_T}(T)$ for any $T \in S_{\mathbb{L}(\mathbb{X}, \mathbb{Y})}$. The final remark states that the order problem for operators between finite-dimensional polyhedral Banach spaces is thereby completely solvable in a computationally effective way.

Concrete consequence for $\ell_\infty^n \rightarrow \ell_1^n$

Let $T \in S_{\mathbb{L}(\ell_\infty^n, \ell_1^n)}$ over the real field. Put

$$R = M_T \cap \text{Ext}(B_{\ell_\infty^n}) = \{\sigma \in \{\pm 1\}^n : \|T\sigma\|_1 = 1\}.$$

Choose a basis $v_1, \dots, v_r$ of $\text{span } R$. For $y \in S_{\ell_1^n}$, the extreme support functionals in $(\ell_1^n)^* = \ell_\infty^n$ are the sign vectors

$$\text{Ext } J(y) = \{\varepsilon \in \{\pm 1\}^n : \varepsilon_i = \text{sign}(y_i) \text{ whenever } y_i \neq 0\}.$$

Let $\eta_1, \dots, \eta_s$ be a basis for the span of all $\text{Ext } J(Tv)$, $v \in R$. For every $v \in R$ and $\varepsilon \in \text{Ext } J(Tv)$, write

$$v = \sum_{i=1}^r \alpha_i v_i, \quad \varepsilon = \sum_{j=1}^s \beta_j \eta_j,$$

and form the coefficient vector

$$z(v, \varepsilon) = (\alpha_i \beta_j)_{1 \leq i \leq r, 1 \leq j \leq s} \in \mathbb{R}^{rs}.$$

Then the Sain–Sohel–Paul index specializes to

$$i_{M_T}(T) = \text{rank span}\{z(v, \varepsilon) : v \in R, \varepsilon \in \text{Ext } J(Tv)\}.$$

Consequently,

$$T \text{ is } k\text{-smooth} \iff \text{rank span}\{z(v, \varepsilon)\} = k.$$

This gives the necessary and sufficient condition requested in the source’s suggested ℓ_∞^n -to- ℓ_1^n case.

Scope limitation

The broad question for arbitrary Banach spaces remains outside this packet. The present identification is a scoped literature implication: the supporting paper cites arXiv:2007.12628 and solves the finite polyhedral operator-order problem, but it does not present itself as a full solution of the final arbitrary-space question from the source paper.

References

- [1] A. Mal, S. Dey, and K. Paul, *Characterization of k -smoothness of operators defined between infinite-dimensional spaces*, arXiv:2007.12628; Linear Multilinear Algebra 70 (2022), no. 18, 3477–3489.
- [2] D. Sain, S. Soheli, and K. Paul, *On k -smoothness of operators between Banach spaces*, arXiv:2407.19455; Bull. Sci. Math. 196 (2024), 103486.